

Design and Experimental Evaluation of an Arduino-Based Battery Thermal Management System with Dual-Sensor Monitoring and Multi-Stage Control

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ABSTRACT

Lithium-ion battery temperature influences performance, degradation, and safety, motivating low-cost supervisory thermal control for small battery-powered systems. This paper presents a bench-scale Arduino-based battery thermal management prototype integrating a DS18B20 contact sensor, an MLX90614 infrared sensor, forced-air cooling, an audible warning, relay-based load shedding, and local LCD monitoring. Valid readings from the two sensors are averaged to obtain an effective temperature, while single-sensor faults are handled by continued operation on the surviving sensor. The implemented controller activates an alarm at 32°C, the cooling fan at 35°C, and disconnects the load at 40°C; fan and load states reset when temperature falls to 32°C or below. In the recorded 40-min test, effective temperature increased from 25.11°C to 40.41°C and then decreased to 31.74°C by 34 min after staged intervention. The fitted heating slope decreased from 0.865°C/min before fan activation to 0.342°C/min while the fan was active, whereas the post-cutoff cooling slope was -1.139°C/min. The two sensors maintained a mean offset of 0.42°C with a standard deviation of 0.04°C. The prototype demonstrates practical embedded sensing, hysteresis, staged actuation, and fault handling, while also illustrating the limitations of simple two-sensor averaging and temperature-only protection.

Keywords: Battery thermal management, lithium-ion battery, Arduino, DS18B20, MLX90614, forced-air cooling, thermal safety

1. Introduction

Lithium-ion batteries are highly temperature sensitive. Elevated temperature accelerates parasitic electrochemical reactions and aging, while strong spatial gradients can produce non-uniform degradation and localized safety risks. Consequently, battery thermal management systems (BTMSs) are designed not only to limit maximum temperature but also to maintain acceptable temperature uniformity and operating efficiency [1], [2], [3]. Air cooling remains attractive in low-cost systems because it is mechanically simple and avoids pumps, coolant leakage, and complex manifolds, although its lower heat-transfer capability limits use at high thermal loads [4].

Thermal safety also depends on sensing quality. Surface temperature can lag internal cell temperature, and a single sensor cannot detect all local hot spots. Recent research therefore increasingly uses distributed temperature sensing and combinations of thermal, electrical, pressure, gas, strain, and acoustic measurements for earlier and more reliable warning [5], [6], [7]. Within this broader context, the present study focuses on a compact educational prototype rather than an automotive-grade battery management system. Its purpose is to demonstrate heterogeneous temperature sensing, staged cooling and protection, hysteresis, sensor-fault handling, and quantitative evaluation of a complete embedded control loop.

2. Literature Review

2.1 Thermal management strategies

BTMS approaches commonly include air, liquid, phase-change material (PCM), heat-pipe, thermoelectric, and hybrid architectures [3], [8], [9]. Liquid cooling generally offers stronger heat rejection but requires pumps, sealed flow paths, and additional control hardware. PCM systems can absorb transient heat passively but may suffer from low thermal conductivity and regeneration limitations. Hybrid arrangements can improve both peak-temperature suppression and temperature uniformity, at the cost of greater system complexity [8], [10].

Forced-air systems remain useful where cost, mass, packaging simplicity, and maintainability dominate. Qin *et al.* identify inlet/outlet placement, channel geometry, airflow rate, flow reversal, and control strategy as key design variables in air-cooled

BTMSs [4]. Mahamud and Park further showed that reciprocating airflow can improve temperature uniformity compared with fixed unidirectional flow [11]. These findings support forced-air cooling for a proof-of-concept platform while also defining the limitations of simple relay-driven on/off cooling.

2.2 Temperature sensing and early warning

The DS18B20 is a contact digital temperature sensor, whereas the MLX90614 estimates object temperature radiometrically. The former therefore responds to conducted heat at its mounting point, while the latter depends on surface emissivity, field of view, and local radiative conditions [12], [13]. Agreement between the two channels can improve plausibility checking and redundancy, but arithmetic averaging does not by itself establish improved absolute accuracy.

Distributed optical sensing has demonstrated the value of resolving internal and surface temperature fields rather than relying on a single monitored location [5]. Dual-temperature sensing has also been explored specifically for in-situ lithium-ion battery monitoring [14]. More recent thermal-runaway warning literature emphasizes multi-parameter fusion because gas evolution, pressure changes, deformation, voltage behavior, and internal temperature may provide useful information before severe surface-temperature escalation [6], [7], [15].

Table 1 Selected literature and design implications for the present system.

Study	Main focus	Relevance to this work
Arora [3]	BTMS review	Establishes cooling-performance versus complexity trade-offs.
Qin <i>et al.</i> [4]	Air cooling	Supports forced-air control while highlighting airflow-design limitations.
Wei <i>et al.</i> [5]	Distributed sensing	Motivates future multi-point temperature monitoring.
Wang <i>et al.</i> [14]	Dual sensing	Supports complementary sensor modalities and redundancy.
Han <i>et al.</i> [6]	Early warning	Shows why temperature-only protection is insufficient for advanced safety systems.

3. Experimental Setup and Control Strategy

3.1 Prototype hardware

The prototype consists of an Arduino Uno, DS18B20 contact temperature sensor, MLX90614 infrared temperature sensor, 16×2 I²C LCD, buzzer, two 5-V relay modules, 12-V cooling

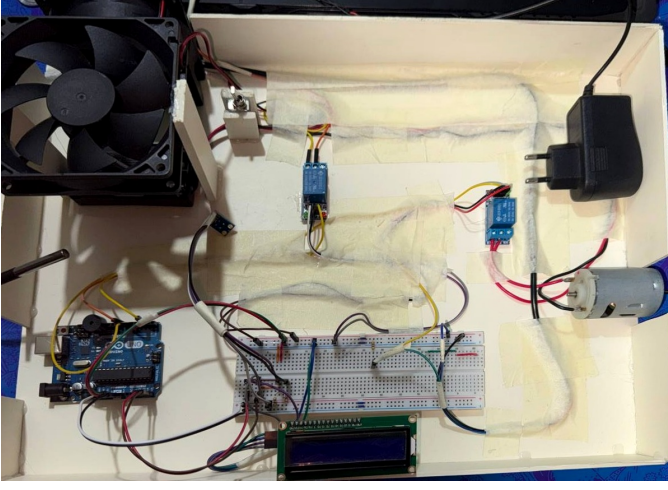


Figure 1 Experimental prototype used for the thermal-control demonstration.

fan, DC motor load, toggle switch, external supply, and two nominal 3.7-V lithium-ion cells. The Arduino performs sensing and supervisory control, while the relays isolate the low-voltage logic from the fan and motor circuits.

The final firmware uses the DS18B20 on D2, fan relay on D6, load relay on D7, and buzzer on D11. The MLX90614 and LCD share the I²C bus on A4/A5. A 4.7-kΩ pull-up is used on the DS18B20 data line. The software rejects clearly invalid sensor values before executing the control logic.

Table 2 Principal hardware interfaces and functions.

Device	Interface	Function
DS18B20	D2	Contact temperature measurement
MLX90614	I ² C	Non-contact object temperature
LCD	I ² C	Temperature and system status
Fan relay	D6	Switches forced-air cooling
Load relay	D7	Disconnects/reconnects motor load
Buzzer	D11	Early-warning alarm

3.2 Sensor fusion and staged control

When both temperature measurements are valid, the effective temperature is computed as

$$T_{\text{eff}} = \frac{T_{\text{DS}} + T_{\text{IR}}}{2}. \quad (1)$$

If one sensor fails, operation continues on the remaining valid sensor and a fault is shown on the LCD. If both sensors fail, the implemented firmware disconnects the load and displays a failure condition.

The control system uses three stages. At $T_{\text{eff}} \geq 32^\circ\text{C}$ the buzzer activates. At $T_{\text{eff}} \geq 35^\circ\text{C}$ the fan is switched on. At $T_{\text{eff}} \geq 40^\circ\text{C}$ the motor load is disconnected. Fan and load states reset only when temperature falls to 32°C or below, producing hysteresis that limits relay chatter around the switching thresholds.

3.3 Experimental procedure

The system was tested from an initial effective temperature of approximately 25°C while the motor load produced heating. DS18B20, MLX90614, and effective temperatures were recorded every two minutes from 0 to 40 min. The logging interval is much slower than the firmware loop, so threshold crossing times are discussed using the nearest recorded points rather than claiming sub-minute timing precision. Three descriptive quantities were extracted: the inter-sensor difference, linear temperature slopes before and after control interventions, and the peak-to-recovery temperature drop. Because only one thermal cycle was available, the results are treated as prototype validation rather than a statistical performance study.

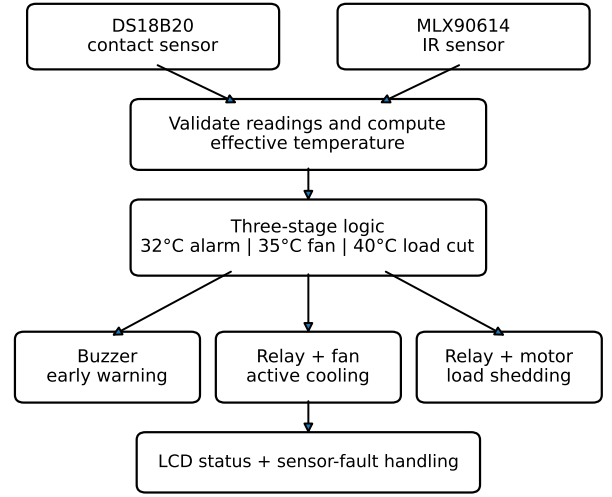


Figure 2 Overall sensing, decision, and actuation flow of the prototype.

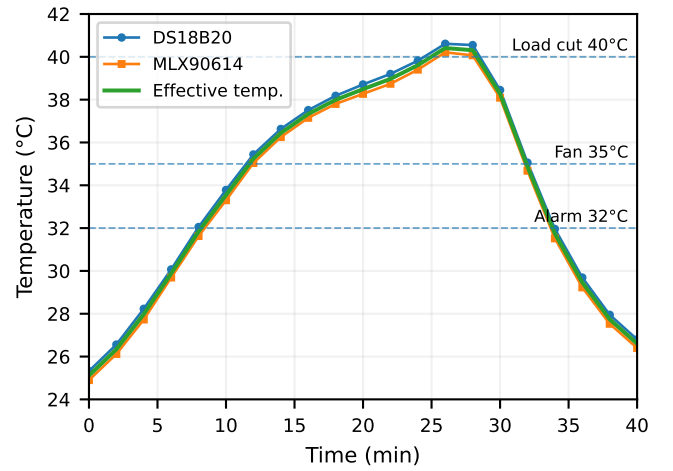


Figure 3 Measured thermal response and implemented control thresholds.

4. Results and Discussion

4.1 Temperature response and staged intervention

Figure 3 shows the measured temperature history. Effective temperature rose from 25.11°C at startup to 31.84°C at 8 min and 33.54°C at 10 min, so the 32°C warning threshold was crossed between these measurements. The effective temperature reached 35.24°C at 12 min, corresponding to fan activation. It continued to rise more slowly until the 40°C protection level was reached, with a recorded peak of 40.41°C at 26 min when the load had been disconnected.

After load removal, temperature remained near the peak at 28 min and then fell to 38.27°C at 30 min, 34.87°C at 32 min, and 31.74°C at 34 min. The resulting peak-to-recovery reduction was 8.67°C in approximately 8 min. Once the effective temperature returned to the 32°C reset region, the fan and alarm were deactivated and the load was allowed to reconnect.

The fitted heating slope before fan activation (0–10 min) was $0.865^\circ\text{C}/\text{min}$. During the fan-active interval (12–26 min), the fitted slope decreased to $0.342^\circ\text{C}/\text{min}$. After load disconnection, the 26–34 min slope became $-1.139^\circ\text{C}/\text{min}$. This change is consistent with the intended staged response. The reduction cannot be attributed exclusively to the fan, however, because no otherwise-identical fan-off control experiment was conducted and natural heat loss also increases as the system becomes hotter.

Table 3 Derived indicators from the recorded thermal cycle.

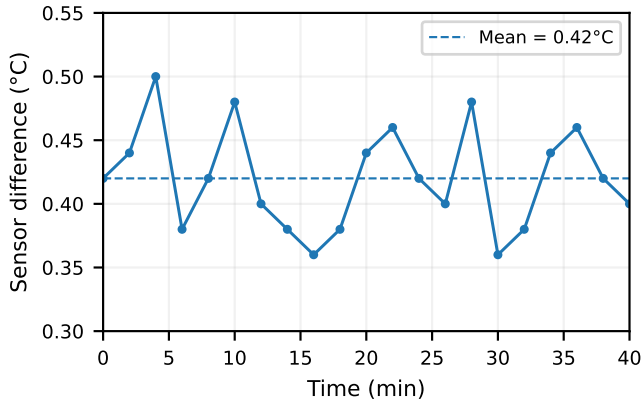


Figure 4 Difference between contact and infrared temperature readings.

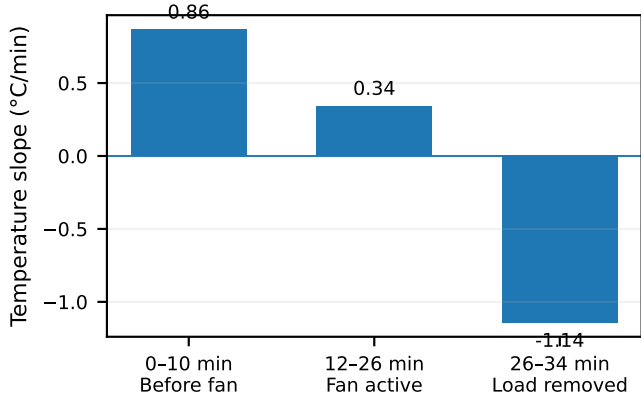


Figure 5 Descriptive temperature slopes before cooling, during fan operation, and after load removal.

Indicator	Value
Initial effective temperature	25.11°C
Peak effective temperature	40.41°C
Peak time	26 min
Mean DS18B20–MLX90614 offset	0.42°C
Offset standard deviation	0.04°C
Heating slope, 0–10 min	0.865°C/min
Heating slope, 12–26 min	0.342°C/min
Cooling slope, 26–34 min	-1.139°C/min
Peak-to-recovery reduction	8.67°C

4.2 Sensor agreement

The DS18B20 reading remained above the MLX90614 throughout the test. The mean difference was 0.42°C with a standard deviation of 0.04°C, indicating a highly repeatable but systematic offset. Figure 4 shows that the offset remained approximately constant as temperature changed. This behavior is reasonable because the two devices use different measurement mechanisms and mounting conditions. The result supports redundancy and consistency checking, but it does not prove that averaging improves absolute accuracy. A calibrated reference thermometer would be required to quantify sensor bias.

4.3 Control effect and engineering interpretation

Figure 5 summarizes the three fitted thermal-response phases. The smaller positive slope after fan activation indicates that the rate of temperature increase was reduced, whereas the negative slope after load removal confirms strong recovery once both heat generation and forced-air cooling acted in the favorable direction.

The selected 32/35/40°C thresholds should be interpreted as conservative supervisory setpoints for a demonstration system, not as universal lithium-ion thermal-runaway limits. Thermal-runaway behavior depends strongly on cell chemistry, state of charge, aging, construction, and abuse mode [6], [7]. Advanced warning systems increasingly combine temperature with gas, pressure, electrical, deformation, and other signals [6], [15].

The present prototype therefore demonstrates staged thermal protection, but not thermal-runaway prediction.

Important limitations remain. Only one thermal cycle was logged, there was no independent calibrated reference sensor, the system monitors one thermal zone, and cooling is binary rather than variable speed. Future versions should use repeated experiments, multi-point contact sensing, current/voltage and state-of-charge monitoring, PWM fan control, formal fault-mode analysis, and hardware rated for the current and isolation requirements of real battery packs.

5. Conclusions

An Arduino-based battery thermal management prototype was developed with heterogeneous dual-temperature sensing, forced-air cooling, audible warning, load shedding, hysteresis, and sensor-fault handling. During the 40-min test, effective temperature increased from 25.11°C to 40.41°C and returned to 31.74°C by 34 min after staged intervention. The heating slope decreased after fan activation, and the post-cutoff response showed a strong negative cooling slope. The two sensors tracked consistently with a stable 0.42°C mean offset, demonstrating repeatability while also revealing the need for calibration before claiming improved measurement accuracy. The prototype provides a compact platform for demonstrating embedded thermal protection and can be extended toward multi-point sensing, variable-speed control, richer battery-state measurements, and multi-parameter safety monitoring.

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